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To Whom It May Concern,

I am pleased to provide this reference for Mr Mingyu Song, who volunteered in my research laboratory at the University of British Columbia from May to August 2026. During this period, Mr Song worked on a technically challenging project focused on using a wearable camera and inertial measurement unit to estimate human movement through simultaneous localization and mapping (SLAM).

The project required Mr Song to engage with a range of computer vision and sensor-fusion methods and tools, including visual-inertial SLAM, ORB-SLAM3, ROS2, OpenCV, and Linux-based computing environments. Their work included camera calibration, integration of camera and IMU data, and implementation and testing of SLAM methods for estimating camera trajectories during movement.

Over the course of the summer, Mr Song made progress toward implementing a functioning visual-inertial tracking system from live-streamed video and inertial data. I observed demonstrations of their work on several occasions, including a live demonstration of the system operating with the wearable camera while moving about our lab space. Developing such a system involves a number of non-trivial technical challenges, and their work provided them with practical experience applying computer vision and sensor-fusion methods to a real research problem.

Mr Song approached an ambitious computational project and gained experience working with tools and methods that extend beyond a typical undergraduate curriculum. Their experience in the lab should provide a useful foundation for further study or technical work in areas such as computer vision, robotics, wearable sensing, and biomedical engineering.

I am pleased to provide this reference in support of Mr Song's future academic and professional opportunities.

Sincerely,

Dr. Jesse Charlton

Assistant Professor

Director, Applied Musculoskeletal Biomechanics Laboratory

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